

The v1→v12 arc: four phases, one variable each

1 · Prefill schedule

v1 naive

HBM-bound; L2 masks it



v2 tiled

up to 3.2x; 0% DRAM cut



v3 online

S gone; 3–7x slower



v4 fused

1.7–2.6x; GEMV wall



v5 WMMA

predicted, not measured



2 · Decode schedule

v6 split-KV

5.7–8.2x; per-CTA



v7 paged

flat %HBM, all batch



v8 GQA pack

8.6x at G=8



v8.7 score-st.

1.1–1.6x; recurrence out



3 · Precision

v9 FP8 KV

1.3x; capacity+accuracy



v10 NVFP4

3.56x capacity; no latency



4 · Shape + engine

v11 MLA

M=128; still per-CTA



v12 tcgen05

2.4→1785; work-starvation



roofline landed



partial



missed (per-CTA layer corrected it)